DriverPass Business Requirements

Kyle DeJong

An academic design proposal for a fictional driving education business. These requirements describe the proposed system, its users, constraints and assumptions. Performance and security targets are design requirements, not measured results from an implemented application. The scenario and named stakeholders were supplied for undergraduate coursework.

Purpose

The academic scenario concerns DriverPass, a fictional driving education company owned by Liam, with Ian as the IT officer.

The purpose of the project is to design a web-based system for DriverPass to help customers prepare for driving tests via online education, practice tests, and on-the-road driving lessons.

The system should support both customer self-service and employee-assisted service, to allow customers to manage their own accounts and reservations while also allowing DriverPass employees to assist via phone or in person.

The system should allow customers to purchase DriverPass packages, schedule driving lessons, access the educational materials in their package, and view their progress through practice tests and lessons.

The system should allow DriverPass employees and administrators to manage customer information, reservations, packages, drivers, vehicles, reports, and system access according to their assigned roles.

The system should be cloud-based and accessible through desktop and mobile browsers, so DriverPass can operate the business with minimal local technical overhead.

System Background

DriverPass wants to address a gap between aspiring drivers and DMV testing by giving customers more structured preparation than studying old tests alone.

DriverPass's solution combines online education, online practice tests, in-person DMV rules instruction, and on-the-road driving lessons with trainers.

The system will need to support customer accounts, package selection, payment for packages, access to educational materials, practice test progress, and driving lesson reservations.

The system will need a scheduling component that connects customers, drivers, vehicles, dates, times, lesson duration, and package limits.

DriverPass will need an employee-facing interface for the owner, IT officer, secretary, and other authorized employees to manage accounts, reservations, drivers, vehicles, packages, lesson notes, and customer records.

The system will need role-based access controls so customers, the secretary, the IT officer, and management do not all have the same permissions.

The system will need to support DriverPass's current package structure, including 6-hour, 8-hour, and 12-hour training packages, with each road lesson scheduled in 2-hour blocks.

The system should allow DriverPass to disable packages that are no longer available, while larger package customization or module changes should be treated as a future development concern.

The system will need to receive or access DMV rules, policies, and sample-question updates so DriverPass's training materials remain current.

The system will need reporting and audit tracking so DriverPass can review reservation activity and identify who created, changed, or canceled appointments.

Objectives and Goals

- The completed system will allow customers to create accounts, log in, reset forgotten passwords, and maintain their profile information.
- The completed system will store customer information needed for DriverPass services, including first name, last name, address, city, state, ZIP code, phone number, email address, pickup location, drop-off location, and payment information needed for package registration or purchase.
- The completed system will allow customers to view available DriverPass packages and purchase the package that fits their training needs.
- The completed system will allow DriverPass to disable unavailable packages so customers cannot register for packages that are no longer being offered.
- The completed system will support DriverPass's current package structure, including 6-hour, 8-hour, and 12-hour driving packages.
- The completed system will schedule driving lessons in 2-hour blocks and track the assigned customer, driver, vehicle, date, start time, and end time.
- The completed system will allow customers to create, modify, and cancel driving lesson reservations online through their account.
- The completed system will allow the secretary to create, modify, and cancel driving lesson reservations for customers who call or visit DriverPass directly.
- The completed system will track customer training progress, including online practice test name, time taken, score, and status information.
- The completed system will display practice test statuses clearly, including not taken, in progress, failed, and passed.
- The completed system will store lesson information, including scheduled lessons, completed lessons, lesson times, and driver comments.
- The completed system will allow authorized DriverPass employees to access the customer, reservation, driver, vehicle, package, and lesson information needed for their roles.
- The completed system will give the IT officer administrative access to manage accounts, reset passwords, modify user access, and block access when needed.
- The completed system will log reservation activity, including who created, modified, or canceled an appointment.
- The completed system will generate activity reports so DriverPass can review system changes and identify responsibility if an issue occurs.
- The completed system will receive or access DMV rules, policies, and sample-question updates, and notify DriverPass when those updates are available.
- The completed system will allow authorized users to download reports or selected information for offline review, while preventing offline edits from changing the live system.
- The completed system will provide customer contact features, including a contact page for DriverPass and a way for authorized employees to contact students.

Requirements

Nonfunctional Requirements

Performance Requirements

- The system shall run as a cloud-based web application accessible through current desktop and mobile web browsers.
- As an initial performance target to be confirmed with DriverPass, the system shall load standard customer and employee pages within 5 seconds under normal operating conditions. This includes account pages, reservation pages, package pages, test progress pages, and lesson information pages.
- As an initial performance target to be confirmed with DriverPass, the system shall complete standard reservation actions within 5 seconds under normal operating conditions. This includes creating, modifying, or canceling a driving lesson reservation.
- The system shall process package purchases and return confirmation of successful payment within 10 seconds after the approved payment-processing service responds.
- The system shall generate standard activity reports within 10 seconds under normal operating conditions.
- The system shall allow authorized users to download standard reports or selected information for offline review within 30 seconds. Larger exports may require additional processing time.
- As an initial capacity target to be confirmed with DriverPass, the system shall support at least 25 concurrent customer sessions and 5 concurrent employee or administrator sessions during initial deployment.
- The system shall check for DMV rules, policies, or sample-question updates at least once per day and notify authorized DriverPass users when an update is available.

Platform Constraints

- The system shall be operating-system agnostic for end users, meaning customers and employees should be able to access the system from supported browsers on Windows, macOS, iOS, Android, or similar current operating systems.
- The system shall be built as a cloud-hosted web application rather than a locally installed desktop application.
- The system shall use a centralized back-end database to store customer accounts, reservations, packages, drivers, vehicles, lesson records, test progress, audit logs, and report data.
- The system shall not allow offline changes to live system data. Offline access shall be limited to downloaded reports or exported information.
- The system shall support downloadable report formats that can be reviewed outside the system, such as spreadsheet-compatible exports.
- The system shall integrate with an external payment-processing service for package purchases, rather than requiring DriverPass to directly manage payment processing infrastructure.
- The system shall support an external DMV update connection or update feed so DriverPass can receive current rules, policies, and sample questions.
- The cloud hosting environment shall provide backup and security support so DriverPass does not need to manage those services through local infrastructure.
- The system shall require an active internet connection for account access, reservation changes, payment processing, DMV updates, and employee administration.

Accuracy and Precision

- The system shall distinguish users by assigned role, including customer, secretary, IT officer, owner/manager, and other authorized employee roles.

- The system shall require each user account to have a unique login identifier so customer and employee actions can be accurately tied to the correct account.
- Usernames or email addresses shall not be case-sensitive for login purposes, but passwords shall be case-sensitive.
- The system shall validate required customer information before account creation or reservation scheduling is completed, including first name, last name, address, city, state, ZIP code, phone number, email address, pickup location, drop-off location, and selected package.
- The system shall validate driving lesson reservations before saving them. A valid reservation must include the customer, assigned driver, assigned vehicle, date, start time, end time, and selected package.
- The system shall validate that the customer's drop-off location matches the pickup location unless DriverPass approves a different business rule.
- The system shall schedule driving lessons only in 2-hour blocks.
- The system shall prevent a customer from scheduling more driving lesson hours than their purchased package allows.
- The system shall prevent double-booking of the same driver, vehicle, or time slot.
- The system shall limit practice test status values to not taken, in progress, failed, or passed.
- The system shall accurately store customer test progress, including test name, time taken, score, and status.
- The system shall accurately store lesson information, including lesson time, start hour, end hour, assigned driver, and driver comments.
- The system shall notify the IT officer or authorized administrator when a system conflict prevents a reservation from being saved, when DMV update information cannot be received, or when required customer/reservation data is missing or invalid.

Adaptability

- The system shall allow the IT officer to add, modify, disable, or block user accounts without requiring code changes.
- The system shall allow authorized employees to update customer profile information, reservation information, lesson notes, driver assignments, vehicle assignments, and contact information without requiring code changes.
- The system shall allow DriverPass to disable existing packages when they are no longer available for purchase.
- The system shall not require the initial release to allow nondeveloper users to create entirely new package modules or remove system modules. Larger package customization should be treated as a future development concern.
- The system shall allow authorized DriverPass users to update training content, practice test material, and DMV-related information when new rules, policies, or sample questions are received.
- The system shall be able to receive or access DMV updates without requiring a full system redesign.
- The system shall support routine browser, cloud-hosting, and security updates without requiring major changes to the user-facing interface.
- The system shall support future expansion of package options, training materials, report types, and user roles through later development releases.

Security

- The system shall require each customer and employee to log in with a unique account before accessing account information, reservations, reports, lesson records, or administrative tools.
- The system shall enforce role-based access control so customers, the secretary, the IT officer, owner/manager, and other authorized employees can only access the functions and records appropriate to their role.

- The system shall allow customers to reset forgotten passwords through an automated password reset process.
- The system shall allow the IT officer to reset user accounts, modify user access, and block accounts when an employee leaves DriverPass or when access needs to be restricted.
- As an initial security target to be confirmed with DriverPass, the system shall temporarily lock an account after 5 failed login attempts within 15 minutes.
- The system shall notify the IT officer or authorized administrator when repeated failed login attempts suggest a possible brute-force attempt.
- The system shall use encrypted connections for data exchanged between the user's browser and the cloud-hosted system.
- The system shall protect customer personal information, reservation records, payment-related information, test progress, lesson notes, and driver comments from unauthorized access.
- The system shall process package payments through an approved payment-processing service.
- The system shall not store credit card security codes after payment authorization.
- The system shall maintain audit logs showing who created, modified, or canceled reservations and who made relevant account or system changes.
- The system shall restrict report downloads to authorized DriverPass users.
- The system shall prevent offline users from modifying live system data, so downloaded reports or exported information cannot create duplicate or conflicting records in the active system.

Functional Requirements

- The system shall allow customers to create an account.
- The system shall allow customers to log in to their account.
- The system shall allow customers to reset a forgotten password through an automated password reset process.
- The system shall allow customers to view and update their personal profile information, including name, address, city, state, ZIP code, phone number, email address, pickup location, and drop-off location.
- The system shall allow customers to view available DriverPass packages.
- The system shall allow customers to purchase a DriverPass package through the web interface.
- The system shall process package payments through an approved payment-processing service.
- The system shall allow DriverPass to disable unavailable packages so customers cannot purchase packages that are no longer offered.
- The system shall allow customers to schedule, modify, and cancel driving lesson reservations online.
- The system shall allow the secretary to create, modify, and cancel driving lesson reservations for customers who call or visit DriverPass directly.
- The system shall record each reservation by customer, assigned driver, assigned vehicle, date, start time, end time, and selected package.
- The system shall allow authorized employees to manage driver and vehicle information needed for lesson scheduling.
- The system shall allow customers to access the educational material included in their purchased package.
- The system shall support the in-person DMV rules lesson included with Package Two and Package Three.
- The system shall allow customers with Package Three to access the online class and practice tests.
- The system shall display customer practice test progress, including test name, time taken, score, and status.
- The system shall display practice test status values, including not taken, in progress, failed, and passed.
- The system shall store lesson records, including scheduled lessons, completed lessons, lesson duration, start time, end time, and driver comments.

- The system shall allow authorized employees to view and update customer records, reservation records, lesson notes, package information, driver information, and vehicle information according to their assigned role.
- The system shall allow the IT officer to manage user accounts, reset accounts, modify access, and block access when needed.
- The system shall record reservation activity, including who created, modified, or canceled a reservation.
- The system shall generate activity reports showing relevant system changes and reservation activity.
- The system shall allow authorized DriverPass users to download reports or selected information for offline review.
- The system shall receive or access updates to DMV rules, policies, and sample questions.
- The system shall notify authorized DriverPass users when DMV rules, policies, or sample questions are updated.
- The system shall provide a contact page for customers to contact DriverPass.
- The system shall provide a way for authorized DriverPass employees to contact students.
- The system shall allow assigned drivers or authorized employees to enter driver comments for completed lessons.

User Interface

- The user interface shall be browser-based and accessible from supported desktop and mobile web browsers.
- The customer interface shall allow customers to create an account, log in, reset a forgotten password, view/update profile information, purchase packages, schedule lessons, modify or cancel reservations, access eligible training materials, view practice test progress, and review lesson information.
- The secretary interface shall allow the secretary to enter or update student information and create, modify, or cancel driving lesson reservations for customers who call or visit DriverPass directly.
- The IT officer interface shall allow account management, password/account reset, access modification, and account blocking.
- The owner/manager interface shall allow access to activity reports, reservation history, package information, customer records, and relevant operational data.
- Customer profile pages shall include the customer information required for DriverPass services, including first name, last name, address, city, state, ZIP code, phone number, email address, pickup location, drop-off location, and payment information needed for package purchase.
- The interface shall support additional profile fields including special needs, student photo, and driver photo where applicable.
- The reservation interface shall show the customer, assigned driver, assigned vehicle, date, lesson start time, lesson end time, and selected package.
- The package interface shall show available DriverPass packages and prevent disabled packages from being selected for new purchases.
- The online test progress interface shall display test name, time taken, score, and status.
- The test status display shall use clear status values, including not taken, in progress, failed, and passed.
- The driver notes interface shall display lesson time, start hour, end hour, and driver comments.
- The interface shall include a contact page for customers to contact DriverPass.
- The interface shall provide authorized employees with a way to contact students.
- The interface shall provide report download options for authorized users, while clearly separating downloaded/offline review from live system editing.
- The interface shall present different options based on user role so customers, the secretary, the IT officer, and management do not see the same controls unless their role permits it.

- The driver/trainer interface shall allow assigned drivers to view their scheduled lessons and enter lesson comments where permitted by their role.

Assumptions

DriverPass customers are assumed to have access to a supported desktop or mobile device with an internet connection when using the online customer features.

DriverPass employees are assumed to have access to supported devices and internet connections when using the employee-facing system.

DriverPass is assumed to provide accurate business information needed to configure the system, including package details, lesson lengths, pricing, driver information, vehicle information, and employee role assignments.

DriverPass is assumed to identify which users should have customer, secretary, IT officer, owner/manager, or other employee permissions before the system is deployed.

DriverPass is assumed to select or approve a payment-processing service before payment functionality is finalized.

The DMV is assumed to provide some reliable method for DriverPass to receive or access updated rules, policies, and sample questions.

The cloud hosting provider is assumed to provide the backup and security support needed for DriverPass to avoid managing those services locally.

Limitations

The initial system release will not allow nondeveloper users to create entirely new package modules or remove system modules. DriverPass will be able to disable existing packages, but larger package customization should be handled as a future development release.

The system will not allow offline editing of live DriverPass data. Offline access will be limited to downloaded reports or exported information for review.

The initial system release will be browser-based and will not include a separate native mobile application.

The system's DMV update feature may be limited by the type of update access the DMV provides. If the DMV does not provide a usable update feed, API, or reliable notification method, additional manual update procedures or future development may be needed.

The payment feature will depend on the selected payment-processing service. Changes to the processor after design approval may require additional integration work.

The project schedule assumes that the major requirements are approved during the customer approval period. Significant scope changes after approval may delay interface development, database work, business logic, testing, delivery, or sign-off.

The initial release will focus on the features identified in the scenario interview. Future features not defined by DriverPass during requirements gathering should be treated as later enhancements.

Project schedule

The planning chart shows the scenario schedule and fictional personnel assignments. A full-resolution copy is available in diagrams/project-schedule.png.

Color	Category/Personnel	Resource Assignment / Tasks
Blue	Jennifer and Sam / Project analysis team	Collect Requirements, Create Use Case Diagrams, Build Activity Diagrams for Each Use Case
Green	Toni and Clark	Research User Interface Designs
Yellow	John	Build Class Diagram
Red	DriverPass / Customer review	Get Customer Approval
Purple	Development team (No individuals specified in transcript)	Build Interface, Link DB to Interface, Build Business Logic, Test System
Grey	Project closeout (No individuals specified in transcript)	Deliver System, Sign-off Meeting

